THE EXACT LOGICAL ERROR PROBABILITY OF THE ROTATED SURFACE CODE UNDER A LOOKUP-TABLE DECODER: SYNDROME-SPACE CHARACTER SUMS COLLAPSE A CERTIFIED BRACKET TO A SINGLE RATIONAL, REPLAYABLE IN THE STANDARD LIBRARY

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Abstract. Part H of this program [PartH] published machine-checkable two-sided rational brackets $L \le P_L \le U$ on the logical error probability of the distance-3 and distance-5 rotated surface codes, one round, circuit-level depolarizing noise, under a fixed lookup-table (coset-leader) decoder — and named its own frontier: exact re-verification of the uncorrectable-set counts at weight 7 exceeds a pure-Python checker, because the shipped method enumerates $\binom{77}{7} = 2{,}404{,}808{,}340$ fault sets. This note removes both the truncation and the wall at $d \le 5$. A syndrome-space character sum (a MacWilliams-type Walsh–Hadamard argument) yields *every* uncorrectable count A_w, $w = 0, \ldots, m$, in a single $O(2^n n)$ pass over the 2^n syndromes — no per-weight enumeration — and its probability-weighted variant yields the *exact* rational P_L, eliminating the bracket altogether. At $d = 5$ this delivers the previously unreachable $A_7 = 832{,}441{,}445$, the full spectrum through $A_{77} = 1$, the undetectable-logical spectrum N_w (re-deriving the circuit-level distance 5), and exact dyadic rationals P_L at both published operating points, each lying strictly inside the corresponding Part H bracket — at 0.36 and 0.51 of the bracket widths, never near an endpoint. A standard-library Python checker re-derives all of it from the mechanism list alone in about two minutes at $d = 5$ within 0.85 GB, and rejects eight classes of tampered certificate. The honest trade: the $\binom{m}{w}$ enumeration wall is exchanged for a 2^n syndrome-space wall, so one-round distance-7 codes ($n \gtrsim 48$ detectors) remain out of reach on a laptop by this method, and we state precisely why. The exact values supersede the Part H brackets; the brackets survive as the independent check the exact values passed.

1. The question

Part H [PartH] asked whether the logical error probability P_L of a quantum error-correcting code, in the deep sub-threshold regime that defeats direct Monte Carlo sampling [MWB], can be pinned by a certificate a skeptic replays with nothing but a standard-library Python interpreter. Its answer was a two-sided exact-rational bracket $L \le P_L \le U$, built from exhaustively enumerated uncorrectable-set counts A_w up to a truncation weight $W_{\max}$, with the width $U - L$ a Poisson-binomial tail. Its §6 then located the method's own frontier, and its Question 7.1 asked:

> *Is there a proof-carrying counter for the exact weighted enumeration at $W_{\max} = 7$ — a CPOG-style weighted model-counting certificate, or an exact locality-exploiting*

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Computation and authorship. All derivation, implementation, verification, and drafting in this work were produced by **Claude** (Anthropic), directed by the author, on a single Apple laptop. The public artifact is a certificate together with a checker that imports only the Python standard library and shares no code with the engine that produced the certificate; the checker re-derives every certified quantity from the mechanism list alone. The detector error models are those of Part H [PartH], generated there with Stim 1.16.0 (Gidney), which is a generator and is *not* in the trusted base (§5).

Prior-art record. Novelty was swept on August 12, 2026 against arXiv and the web; the dated record ships with the artifacts as `SWEEP-RECORD-WEDGE2-2026-08-12.md`. The character-sum technique is MacWilliams-type and classical in lineage; the claim is scoped to the intersection stated in §3.

convolution / transfer-matrix — that a standard-library checker can re-verify in minutes rather than hours?

This note answers that question affirmatively at $d \le 5$, and with a stronger object than the question requested. The answer is not a faster enumeration and not a tighter bracket: a character sum over the syndrome space computes all counts A_w simultaneously, with cost independent of the weight, and its probability-weighted form computes P_L *exactly* — as a single reduced rational — so that the truncation weight, the tail, and the bracket disappear from the certified statement. The Part H brackets are retained by the new certificates in one role only: as published, independent enclosures that the exact values are verified to satisfy. Everything else about the Part H trust architecture — the mechanism list as trust root, the fixed decoder, the standard-library checker, the private generator — is unchanged, and the new checker re-verifies in two minutes, at all weights at once, what the enumeration checker could not reach at weight 7 in hours.

2. The identity

Model. Fix a detector error model (DEM): n detectors, one logical observable, m independent fault mechanisms; mechanism j carries a detector mask $d_j \in \mathbb{F}_2^n$, an observable bit $o_j \in \mathbb{F}_2$, and a probability $p_j \in (0,1)$, an exact dyadic rational (the IEEE-754 double emitted by Stim). Write $v_j = (d_j, o_j) \in \mathbb{F}_2^{n+1}$. A fault configuration $F \subseteq [m]$ has syndrome $\sigma(F) = \bigoplus_{j\in F} d_j$, observable flip $\mathrm{obs}(F) = \bigoplus_{j\in F} o_j$, weight $|F|$, and probability $\mathbb{P}(F) = \prod_{j\in F} p_j \prod_{j\notin F}(1-p_j)$.

Decoder. The decoder $D : \mathbb{F}_2^n \to \mathbb{F}_2$ is the coset-leader lookup table of Part H: a full breadth-first search over the syndrome space from the zero syndrome, expanding by the mechanism masks in canonical index order, first assignment winning; $D(s)$ is the observable flip of the first-found minimum-cardinality configuration reaching s. In the two DEMs certified here the masks span $\mathbb{F}_2^n$, so every syndrome is assigned (the checker verifies this), and F is uncorrectable iff $D(\sigma(F)) \ne \mathrm{obs}(F)$. On syndromes of BFS depth $\le W_{\max}$ this table coincides with the depth-truncated table of Part H, so the decoder is the same decoder.

The counts, all at once. For $u \in \mathbb{F}_2^{n+1}$ let $I(u)$ be the failure indicator: $I(s,b) = 1$ iff $b \ne D(s)$. Let A_w be the number of uncorrectable F with $|F| = w$, and let $N_w = \#\{F : |F| = w,\ \sigma(F) = 0,\ \mathrm{obs}(F) = 1\}$ count the undetectable logical operators, so that the circuit-level distance is $\min\{w : N_w > 0\}$. For $y \in \mathbb{F}_2^{n+1}$ write $\langle y, u\rangle$ for the standard pairing, $\widehat{f}(y) = \sum_u (-1)^{\langle y,u\rangle} f(u)$ for the (unnormalised) Walsh–Hadamard transform, and

$$a(y) \;=\; \#\{\, j : \langle y, v_j\rangle = 0 \,\}, \qquad g_w(k) \;=\; [t^w]\,(1+t)^k(1-t)^{m-k}.$$

Theorem 2.1 (Counting identity)**.** $A_w \;=\; \dfrac{1}{2^{\,n+1}} \displaystyle\sum_{y\in\mathbb{F}_2^{n+1}} \widehat{I}(y)\, g_w(a(y))$, *and the same identity with $\widehat{I}$ replaced by the transform of the indicator of $(0,1)$-cosets gives* $N_w = \dfrac{1}{2^{\,n+1}} \displaystyle\sum_{y\in\mathbb{F}_2^{n+1}} (-1)^{y_b}\, g_w(a(y))$, *where $y = (y_s, y_b)$, $y_s \in \mathbb{F}_2^n$, $y_b \in \mathbb{F}_2$.*

Proof. Let $N_w(u) = \#\{F : |F| = w,\ \bigoplus_{j\in F} v_j = u\}$. In the group algebra of $\mathbb{F}_2^{n+1}$, $\sum_{w,u} N_w(u)\, t^w e_u = \prod_j (1 + t\, e_{v_j})$, and the character $\chi_y(e_u) = (-1)^{\langle y,u\rangle}$ maps the product to $\prod_j (1 + t(-1)^{\langle y,v_j\rangle}) = (1+t)^{a(y)}(1-t)^{m-a(y)}$. Fourier inversion gives $N_w(u) = 2^{-(n+1)} \sum_y (-1)^{\langle y,u\rangle} g_w(a(y))$, and $A_w = \sum_u I(u) N_w(u) = 2^{-(n+1)} \sum_y \widehat{I}(y)\, g_w(a(y))$ by exchanging the finite sums. Taking $u = (0,1)$ instead of summing against I gives the N_w statement, since $(-1)^{\langle y,(0,1)\rangle} = (-1)^{y_b}$. $\square$

Theorem 2.2 (Exact probability identity)**.** $P_L \;=\; \dfrac{1}{2^{\,n+1}} \displaystyle\sum_{y\in\mathbb{F}_2^{n+1}} \widehat{I}(y) \prod_{j\,:\,\langle y,v_j\rangle=1} (1-2p_j).$

Proof. Identical character argument applied to $\prod_j\big((1-p_j)\,e_0 + p_j\,e_{v_j}\big)$: the coefficient of e_u is the total probability of $\{F : \bigoplus_{j\in F} v_j = u\}$, and χ_y maps each factor to $(1-p_j) + p_j(-1)^{\langle y,v_j\rangle}$, which is 1 when $\langle y, v_j\rangle = 0$ and $1-2p_j$ otherwise. Summing the inverted coefficients against I gives $P_L = \sum_u I(u)\cdot[\text{coefficient}]$ as claimed. ☐

Splitting $y = (y_s, y_b)$ makes both identities computable by transforms of n-bit arrays: with $c(s) = 1$ (all syndromes reachable), $e(s) = (-1)^{D(s)}$, one finds $\widehat{I}(y_s, 0) = 2^n[y_s = 0]$ and $\widehat{I}(y_s, 1) = -\widehat{e}(y_s)$; and $a(y)$ becomes $a_0(y_s) = \#\{j : \langle y_s, d_j\rangle = 0\}$ resp. $a_1(y_s) = \#\{j : \langle y_s, d_j\rangle = o_j\}$, each obtainable for all y_s from one transform of a sparse mask histogram. Four transforms of length 2^n therefore produce every A_w and every N_w; the total cost is $O(2^n n)$ arithmetic operations on integers of magnitude at most 2^n, plus the $O(2^n m)$ BFS that builds D — *independent of* w, which is what removes the $\binom{m}{w}$ wall. For Theorem 2.2 the product $\prod_{\langle y,v_j\rangle=1}(1-2p_j)$ depends only on how many mechanisms of each distinct probability class have $\langle y, v_j\rangle = 1$ (17 classes at $d = 5$, 14 at $d = 3$), so the exact big-integer assembly groups the 2^n characters by a class-count key; all denominators are powers of two, so the common-denominator scaling is a bit shift and the result is a dyadic rational, reduced at the end. Cancellation is severe — the sum is 2^n minus a nearly-equal character sum — which is exactly why the arithmetic is exact integers throughout; a floating-point evaluation of Theorem 2.2 would be meaningless at these magnitudes.

Two closed-form invariants gate the whole computation.

Proposition 2.3. *If some* F_0 *satisfies* $\sigma(F_0) = 0$, $\mathrm{obs}(F_0) = 1$ *(true here:* $N_5 = 111 > 0$ *at* $d = 5$, $N_3 = 9 > 0$ *at* $d = 3$*), then* $\sum_w A_w = 2^{m-1}$. *If moreover the* (d_j, o_j) *span* $\mathbb{F}_2^{n+1}$ *(true here), then* $\sum_w N_w = 2^{m-n-1}$.

Proof. $F \mapsto F \triangle F_0$ is an involution on subsets fixing σ and flipping obs; since $D(\sigma)$ is fixed, exactly one of each pair is uncorrectable, giving $2^m/2$. The second claim: $F \mapsto (\sigma(F), \mathrm{obs}(F))$ is a surjective homomorphism $(\mathbb{F}_2^m, \triangle) \to \mathbb{F}_2^{n+1}$, so each fiber has $2^{m-(n+1)}$ elements. ☐

3. Where this sits

The character sum is not new mathematics: computing the probability of correct syndrome decoding from coset weight enumerators via MacWilliams-type transforms is classical coding theory [JP]. What the classical setting lacks is everything the certificate needs: there, the channel is symmetric with one parameter and the error positions are the code coordinates; here, $m > n$ heterogeneous dyadic mechanisms carry an extra observable bit, the decoder is a specific tabulated function fixed by a BFS convention, and the output is an exact rational that an independent standard-library checker re-derives, hash-bound to a mechanism list. The neighbouring modern literature splits as in Part H: estimators return values with error bars [MWB, BV]; tensor-network maximum-likelihood computations [BSV, DP] are numerically exact floating-point contractions with no certified output; closed-form and enumerator-based analyses [Regev, FVC] are approximations or code-level bounds, not decoder-specific exact values; recent uses of Walsh–Hadamard relations on DEMs [Willow, CAHR] estimate or reconstruct the model from data rather than certify anything about a decoder; formal verifiers [LeanQEC, VeriQEC] machine-check Boolean error-correction conditions, not probabilities. The dated sweep (2026-08-12) found no prior exact certified logical error probability of a stated decoder on a circuit-level DEM, by transform methods or otherwise, with independent standard-library re-verification; we claim that intersection and nothing broader, and this note supersedes our own Part H bracket [PartH] rather than any external result.

4. Results

The two DEMs are byte-identical to Part H's (same mechanism lists, same SHA-256 hashes): $d = 3$ with $n = 8$, $m = 23$; $d = 5$ with $n = 24$, $m = 77$. All counts below are p-independent; the exact P_L is computed at the two published operating points of each code.

The spectra. At $d = 5$ the transform yields in one pass the full uncorrectable spectrum $A_0, \ldots, A_{77}$, whose head is

$$A_3 = 2728, \quad A_4 = 154{,}394, \quad A_5 = 4{,}057{,}999, \quad A_6 = 67{,}711{,}204, \quad \mathbf{A_7 = 832{,}441{,}445},$$

continuing through $A_{77} = 1$ (the full-support configuration). A_3, A_4, A_5 equal the Part H certified counts; A_6 equals the value of the optional $W_{\max} = 6$ certificate of Part H; A_7 and beyond are new — weight 7 is precisely where Part H's enumeration hit its wall ($\binom{77}{7} \approx 2.4 \times 10^9$ subsets, hours of pure-Python re-verification). The undetectable-logical spectrum begins $N_5 = 111$, $N_6 = 786$, $N_7 = 3706$, re-deriving circuit-level distance 5 by transform rather than enumeration. Both closed-form invariants hold to the digit: $\sum_w A_w = 2^{76}$ and $\sum_w N_w = 2^{52}$ (at $d = 3$: 2^{22} and 2^{14}). At $d = 3$ the head $A_2 = 55$, $A_3 = 690$, $A_4 = 4078$ matches Part H, and $A_5 = 16{,}467$ was confirmed by direct enumeration.

The exact values. Table 1 is the core supersession: the exact P_L at all four published operating points, with the Part H bracket it collapses. Every exact value lies *strictly inside* its bracket — the containment $L < P_L < U$ is verified in exact rational arithmetic by the shipped checker, and had it failed the release would have stopped, since it would have falsified Part H.

Table 1. Exact P_L (decimal shadows of exact dyadic rationals; the certificates carry full numerators and denominators, e.g. 1026/1030 digits at d=5, p=10^{-3}) against the Part H brackets $[L, U]$. "Position" is $(P_L - L)/(U - L)$.

code, p	exact P_L	Part H L	Part H U	position
d=3, 10^{-3}	3.358965645735e-4	3.3589593231e-4	3.3589715988e-4	0.515
d=3, 10^{-2}	2.714610979781e-2	2.7101553370e-2	2.7188139126e-2	0.515
d=5, 10^{-3}	2.613858119750e-5	2.6135024167e-5	2.6144956889e-5	0.358
d=5, 10^{-2}	1.734152068512e-2	1.6118428758e-2	1.9426205101e-2	0.370

Three further consistency checks. The d=5, $p = 10^{-3}$ exact value lies inside the tighter optional $W_{\max}$=6 bracket of Part H, [2.6138502142e−5, 2.6138690261e−5], at position 0.42 — a check about 53× more stringent than the released $W_{\max}$=5 bracket. The exact values at both $d = 5$ points lie inside the 10^7-shot Monte Carlo 95% confidence intervals recorded in Part H. And at $p = 10^{-2}$, $d = 5$ — the regime where Part H's bracket was honestly reported **DEAD**, 20.5× wider than Monte Carlo — the exact value 1.734152e−2 settles what the bracket could not resolve: the bracket is now beaten not by more shots but by eliminating the truncation altogether. The supersession is uniform: where the bracket was LIVE it is replaced by the exact value; where it was DEAD it is replaced by the exact value; nothing bracketed remains.

Independent confirmations (pre-registered standard). Because the headline numbers pass beyond what Part H could re-verify, each was confirmed by at least one computational route sharing no arithmetic with the transform engine. At $d = 3$, *three* independent exact routes agree digit-for-digit on P_L at both operating points: the class-binned character sum (Theorem 2.2), a syndrome-space signed convolution reconstructed over 25-bit primes by the Chinese remainder theorem, and a Gray-code full enumeration of all 2^{23} fault configurations in exact integer arithmetic. At $d = 5$, where full enumeration is impossible (2^{77} configurations), the character-sum values were checked against the independent convolution route modulo 24 distinct 25-bit primes — ≈ 600 bits of agreement — at both operating points, and the count spectrum was checked against every Part

H certified count, the $W_{\max} = 6$ engine value, both invariants of Proposition 2.3, and Part H's Monte Carlo. Before this build, the scoping run had also cross-checked A_7 by a truncated-BFS Monte Carlo estimator ($z = 0.19$).

Certification. The permanent record is four certificate JSONs — the mechanism list and hash (identical to Part H), the full A_w and N_w spectra, the circuit-level distance, both invariants, the exact P_L numerator and denominator, and the cited Part H bracket — together with one standard-library checker, `check_wedge2.py`, that re-derives every certified quantity from the mechanism list alone: it rebuilds the full BFS decoder, verifies spanning, runs the Walsh–Hadamard transforms on `array('q')` buffers, assembles the spectra and the exact rational P_L in big-integer arithmetic, and verifies $L \le P_L \le U$ against the embedded Part H bracket. It imports only `hashlib`, `json`, `sys`, `array`, `fractions`; no signals, subprocesses, network, or wall-clock. Runtime: 0.15 s at $d = 3$; at $d = 5$, 129 s and 0.79–0.84 GB peak resident memory per certificate on the build laptop (measured with `/usr/bin/time -l`) — for comparison, Part H's enumeration checker took a comparable 34 s to certify weights ≤ 5 only, ten-plus minutes for ≤ 6, and hours for ≤ 7; this checker certifies all 78 weights *and* the exact P_L in two minutes. A tamper battery of eight corruptions — a flipped count, a perturbed probability with and without a recomputed hash, an altered P_L digit, a swapped observable bit, an overstated distance, a corrupted N_w, a shifted bracket — is rejected gate-by-gate on the d=3 certificate, where the battery replays in under a second (`tamper_demo_w2.py`; the corruptions are certificate-generic and the script runs against any of the four). The identity itself ships with a standard-library self-test (`identity_selftest.py`) that verifies both theorems against brute force on random small DEMs, including non-spanning mask sets, in exact `Fraction` arithmetic. The transform engine, like Part H's enumeration engine, is private and is not needed to check anything.

5. What is certified, and what is trusted

The trust boundary is unchanged from Part H [PartH], and is restated plainly. **Machine-checked:** everything in §4 — spectra, invariants, distance, the exact P_L, bracket containment — is re-derived by the standard-library checker from the certificate's mechanism list, which is hash-pinned and byte-identical to Part H's. **Trusted:** (1) the binding of that mechanism list to Stim's DEM at the stated p lives in the private generator and cannot be re-checked from the public artifact; (2) the independent-mechanism DEM is Stim's standard approximation of the physical depolarizing circuit, and the certified P_L is the DEM's, not the circuit's — Part H measured the gap at $d = 5$, $p = 10^{-3}$ (a raw-circuit Monte Carlo point estimate about 1.17 standard errors above the DEM bracket) and that caveat carries over verbatim; (3) CPython, the OS, the hardware; (4) SHA-256, for bookkeeping only. One trusted item is *removed*: the depth-truncation soundness argument of Part H (that a depth-$W_{\max}$ BFS agrees with the full table on low-weight syndromes) is no longer needed, because the checker builds the full table.

6. The wall that replaces the wall

Part H's frontier was the enumeration $\binom{m}{w}$; this note's method replaces it with the syndrome space 2^n. The exchange is decisively favourable at $d = 5$ — $2^{24} \approx 1.7 \times 10^7$ versus $\binom{77}{7} \approx 2.4 \times 10^9$, with all weights for free — and unfavourable beyond. Every step of the method — the full BFS decoder, the 2^n-length transforms, the class-binned assembly — materialises arrays of length 2^n. One round of the distance-7 rotated surface code has $n \approx 48$ detectors: $2^{48} \approx 2.8 \times 10^{14}$ syndromes, seven orders of magnitude beyond what fits in laptop memory, before any arithmetic. The obstruction to doing better is specific and worth naming: the masks are local (at $d = 5$, every mechanism flips at most two detectors), so the mask-side transforms would factor beautifully over any tensor decomposition of the syndrome space — but the decoder $D(s)$ is a *global* minimum-cardinality object, defined by a BFS over the whole space, and it is $\widehat{e}$, the transform of $(-1)^{D(s)}$, that the identities cannot do without. A method that reaches $d = 7$ must either localise this global object (a frontier transfer-matrix that carries coset-leader information without materialising 2^n states) or abandon the lookup decoder for one with local structure. We state this as the honest limit: *by the method of this note, on a laptop, $d = 7$ circuit-level is out of reach, and no amount of engineering of the present checker changes that.*

7. Questions

Question 7.1. (Part H's Question 7.2, sharpened.) Is there an exact representation of the coset-leader decoder — or of any explicitly specified decoder of comparable quality — whose failure transform $\widehat{e}$ admits evaluation without materialising 2^n states, opening $d = 7$ at one round to a laptop-checkable exact certificate?

Question 7.2. The signed syndrome measure $q(s) = \sum_{F:\sigma(F)=s} \mathbb{P}(F)(-1)^{\mathrm{obs}(F)}$, computable exactly by the convolution route used here as a cross-check, determines the maximum-likelihood decoder for the observable: $D_{\mathrm{ML}}(s) = [q(s) < 0]$. The identities of §2 apply verbatim to *any* tabulated decoder. Can the exact P_L of the maximum-likelihood decoder — the code's true operational figure, Part H's Question 7.4 — be certified at $d = 5$ by combining the two routes, with the sign tabulation itself made checker-re-derivable?

Question 7.3. Multiple rounds multiply the detector count, so the 2^n wall arrives immediately even at $d = 3$. Is there a round-by-round factorisation of the character sum — the time direction is where the DEM *is* local — that trades 2^n for per-round transforms of the single-round syndrome space?

Question 7.4. Part H's Question 7.3 (certify the DEM-to-circuit gap) is untouched by this note and remains open; the exact DEM value makes it sharper, since the gap is now the only distance between the certificate and the physical circuit.

Acknowledgments

This note upgrades Part H of this program and answers its Question 7.1; the question, the decoder convention, the DEMs, the operating points, and the kill-test record are all Part H's, and the supersession is of our own published bracket, not of any external result. The framing that sub-threshold logical error rates defeat direct sampling is Mullan, Weippert, and Brown's [MWB]. The lookup-table decoder is Tomita and Svore's [TS]. The coset-enumerator lineage of the character sum is classical [JP]. The detector error models were produced with Stim (Craig Gidney), used as an untrusted generator. The computation and drafting were AI-assisted as stated in the first footnote; the note's shape — open with the question, cite the computation once, close with the questions it opens — follows the program's standing constraint.

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